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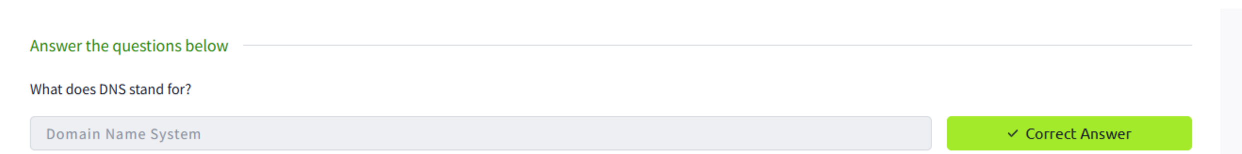
DNS IN DETAIL WITH TRY HACK ME

1.Introduction

DNS- stands for Domain Name System , in this module i have learnt how the DNS changes the domain name to numbers i.e IP addresses that computers can easily interact with. Instead of humans memorizing the IP addresses for google the DNS changes the IP to domain name.

2.What is DNS?

Question
What does DNS stand for?
Answer: **Domain Name System**



3. Domain Hierarchy

Subdomain-I did not know much about subdomains, but with this write up down , i got to understand about the length and the restriction that comes with writing subdomains.A subdomain sits on the left-hand side of the Second-Level Domain using a period to separate it; for example, in the name admin.tryhackme.com the admin part is the subdomain. A subdomain name has the same creation restrictions as a Second-Level Domain, being limited to 63 characters and can only use a-z 0-9 and hyphens (cannot start or end with hyphens or have consecutive hyphens). You can use multiple subdomains split with periods to create longer names, such as jupiter.servers.tryhackme.com. But the length must be kept to 253 characters or less. There is no limit to the number of subdomains you can create for your domain name.

Questions

a)What is the maximum length of a subdomain?
Answer :**63**

Answer the questions below

What is the maximum length of a subdomain?

63

✓ Correct Answer

b) Which of the following characters cannot be used in a subdomain (3 b _ -)?

Answer:_

Explanation: subdomains cannot end or start with a hyphens or have consecutive hyphens

Which of the following characters cannot be used in a subdomain (3 b _ -)?

_

✓ Correct Answer

c)What is the maximum length of a domain name?

Answer:**253**

Explanation:The length of a subdomain must be kept to 253 characters or less

What is the maximum length of a domain name?

253

✓ Correct Answer

d)What type of TLD is .co.uk?

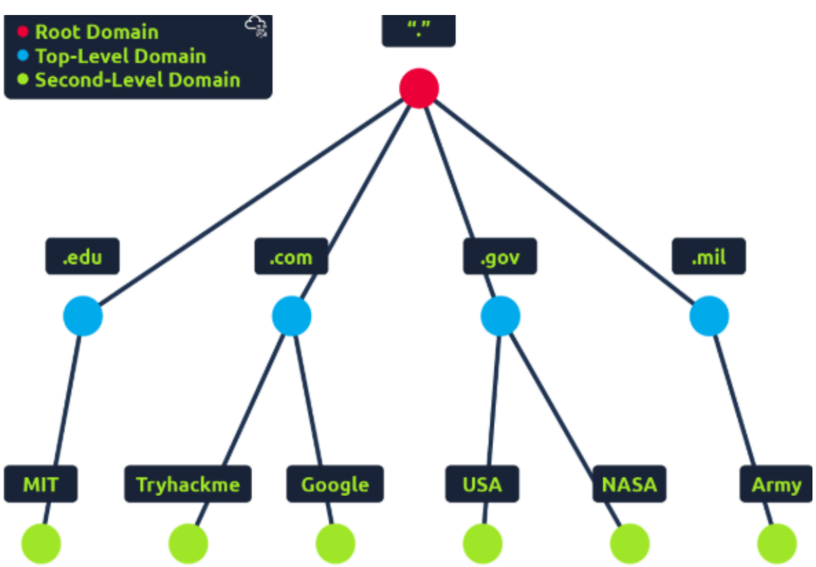
Answer: **ccTLD**

Explanation:ccTLD was used for geographical purposes

What type of TLD is .co.uk?

ccTLD

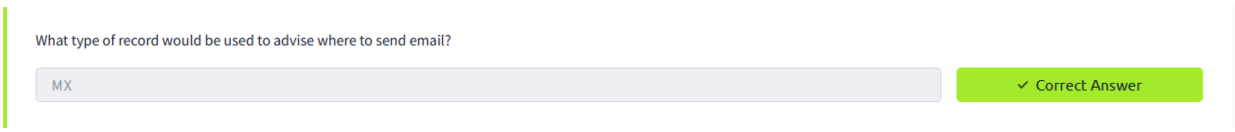
✓ Correct Answer



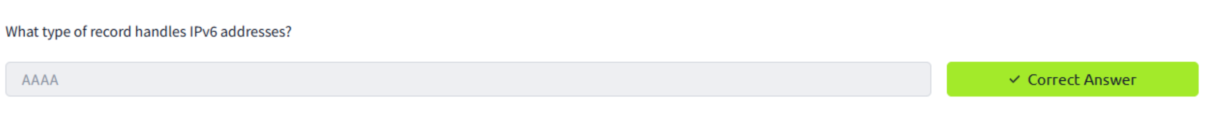
4.Record Types

A Record-These records resolve to IPv4 addresses, for example 104.26.10.229
AAAA Record-These records resolve to IPv6 addresses, for example 2606:4700:20::681a:be5
CNAME Record-These records resolve to another domain name
MX Record-These records resolve to the address of the servers that handle the email for the domain you are querying
TXT Record-TXT records are free text fields where any text-based data can be stored.

Question
a)What type of record would be used to advise where to send email?
Answer: **MX**



b)What type of record handles IPv6 addresses?
Answer:**AAAA**



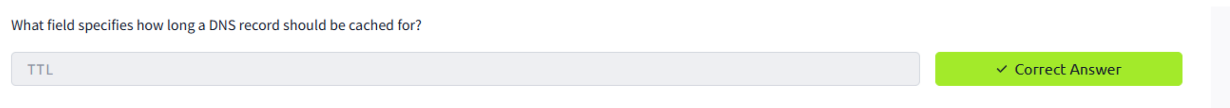
5.Making a request

The DNS resolution process is a hierarchical "search and cache" system designed to translate human-friendly domain names into IP addresses. It begins with a local check of your computer's cache, followed by a query to a Recursive Resolver (usually your ISP), which acts as a middleman to track down the answer. If the resolver lacks a cached copy, it initiates a "relay race" through the internet's infrastructure: starting at the Root Servers to identify the correct Top-Level Domain (TLD) Server (like .com), then moving to the TLD server to find the Authoritative Name Server responsible for that specific domain. This final server provides the actual IP address, which is then passed back down the chain and stored in various caches for a duration set by the Time To Live (TTL) value to speed up future requests.

Question

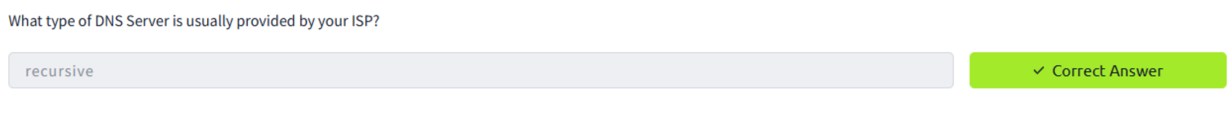
a)What field specifies how long a DNS record should be cached for?

Answer: **TTL**



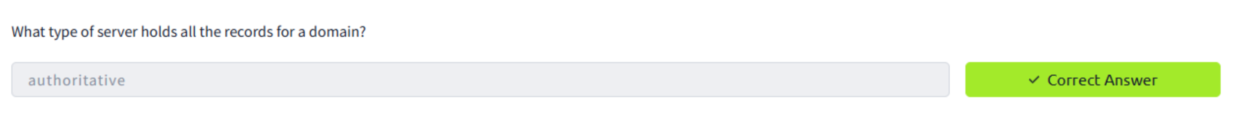
b)What type of DNS Server is usually provided by your ISP?

Answer:**Recursive**



c)What type of server holds all the records for a domain?

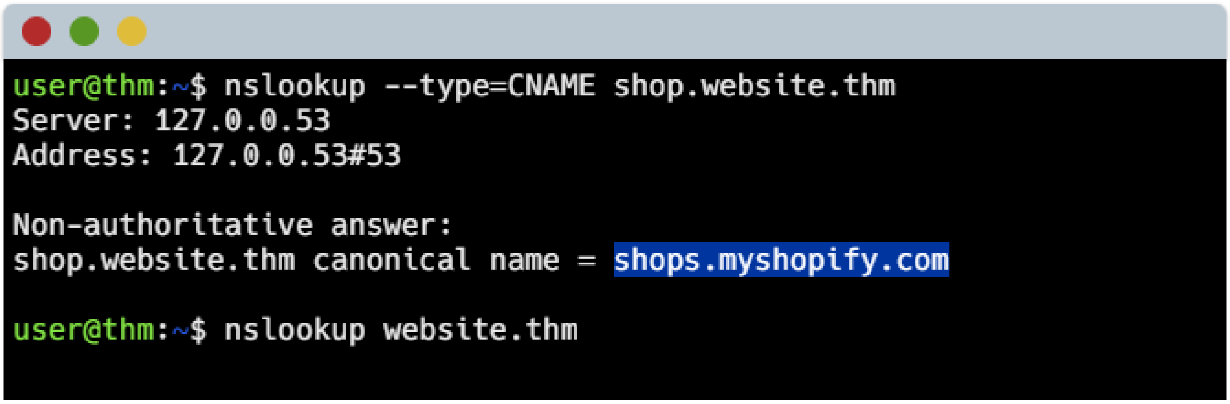
Answer:**authoritative**



6.Practical

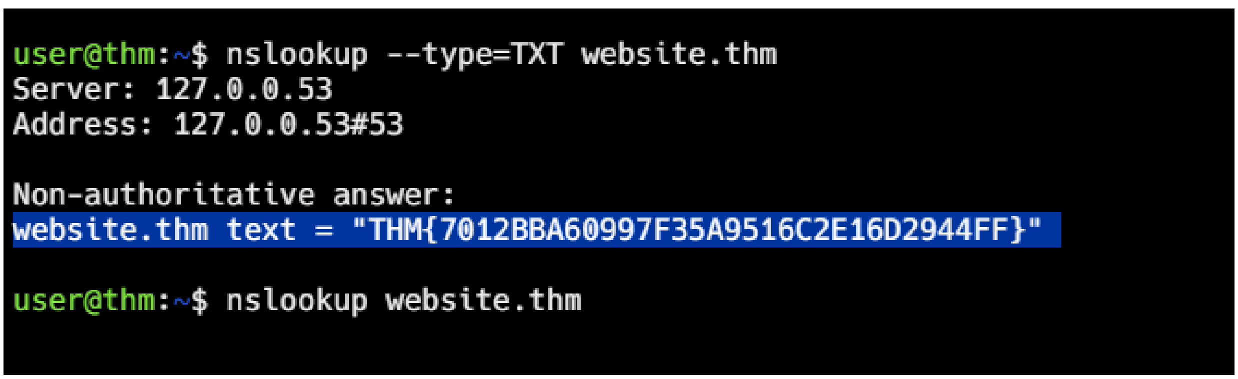
Question

a)What is the CNAME of shop.website.htm?
Answer:**shops.myshopify.com**
Explanation:
After joining the room, I typed shop so that the whole domain would read shop.website.htm. I then changed the DNS type to CNAME.

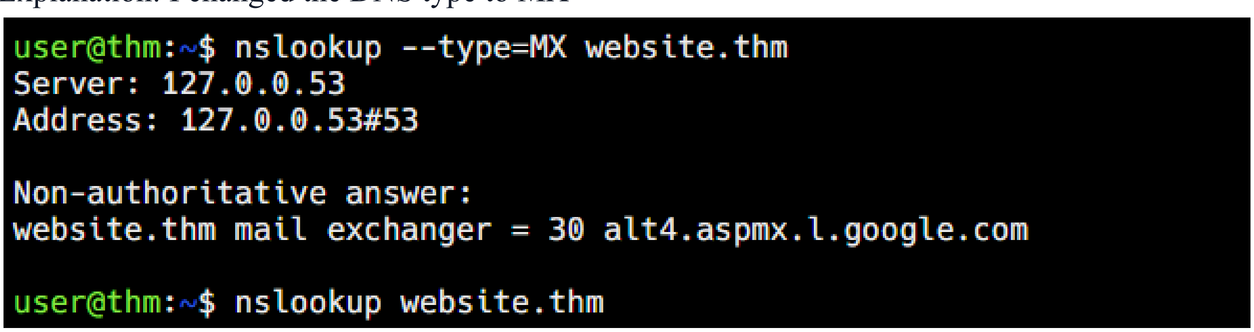


b)What is the value of the TXT record of website.thm?

Answer:**THM{7012BBA60997F35A9516C2E16D2944FF}**
Explanation: I changed the DNS type to TXT



c)What is the numerical priority value for the MX record?
Answer:**30**
Explanation: I changed the DNS type to MX



d)What is the IP address for the A record of www.website.thm?
Answer:**10.10.10.10**

Explanation:I changed the DNS type to A record and got the answer.

```
user@thm:~$ nslookup --type=A website.thm
Server: 127.0.0.53
Address: 127.0.0.53#53

Non-authoritative answer:
Name: website.thm
Address: 10.10.10.10

user@thm:~$ nslookup website.thm
```

7.Conclusion and Summary

I have gained a comprehensive understanding of the DNS hierarchy and the functionality of different DNS records (such as A, MX, and TXT). This expertise is vital for a security analyst, as DNS enumeration is a key phase in penetration testing for mapping a target's infrastructure.

Below is a link of my achievement.
https://tryhackme.com/p/marynyambura657

https://tryhackme.com/room/dnsindetail?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&sharerId=6969d963f9c9a9590800e42d